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WHAT'S NEW IN LIBRARIES

Using virtual and augmented reality in the library

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Abstract

Purpose – The purpose of this paper is to describe the background for libraries to consider the benefit of implementing virtual reality (VR) or augmented reality (AR) usage.

Design/methodology/approach – This paper includes literature review and commentary on this topic that has been addressed by professionals, researchers and practitioners.

Findings – VR or AR are not gimmicks, and must be seen as valid additions to the toolkit that may be used by libraries to engage its audience, not only with the latest technology but also with the goal in mind of ensuring a proper approach to teaching information literacy.

Originality/value – The value in addressing this topic is to examine the notion of virtual or augmented reality and its suggested uses in libraries to support the teaching of information literacy.

Keywords Virtual reality, Libraries, Augmented reality

Paper type Viewpoint

Introduction

Virtual reality (VR) or augmented reality (AR) is a phenomenon that has been in its varying growth stages for years (in the case of VR, decades, as far back as the 1930s). But, today, with a headset available to some and a hand-held device available to many, it benefits library users who are fully expecting such enriched services in the future to include VR or AR:

> […] libraries and institutions of memory have been challenged to find new forms of dialogue with their users and have turned to VR technology to entertain and inform their audience (Angeletaki, 2013).

Affordability, accessibility and a user-friendly platform are all determinant virtues for any technology introduced in the library. To differentiate their definitions:

> Virtual Reality strives to completely simulate an immersive sensory experience indistinguishable from the real […] often using technologies like headsets that attempt to completely deprive human sensory organs like the eyes from any stimuli outside of that which is artificially generated by the VR technology. Augmented Reality strives to overlay additional layers of useful (e.g. actionable, descriptive, informative, directional) media onto the real without replacing it (Zoladz, 2015).

Today, there are even free apps that can be downloaded to one's smart device that can be used to enter the VR/AR ecosystem for educational purposes that "might allow students

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to experience history, math, literature, art or any other area of academics" (Valibrarian, 2015).

Three examples

Example 1 – Aurasma

Aurasma (www.aurasma.com) is an example of a free app that can be downloaded to either an iOS or Android device. It is an easy-to-use AR platform. Upon downloading the app, the user creates original content (called "auras") or chooses from a library of pre-recorded content; selects a specific, static image (called the "target"); and then overlays the aura to the target, and when the user points a device at the target and holds it steady, the aura appears at center screen, thus presenting a recording selected by the originator to add an AR component to the target image.

As an example of its use, Aurasma can bring to life an exhibition in the library's art gallery. A series of posters can be created on a particular theme and auras can then be developed for each poster, bringing a combination of charts, graphs, animations and human-centered narrations to each one, thus providing the gallery viewer with a uniquely enhanced manner of viewing what would have previously been merely a group of flat, wall posters. Several of the auras might contain the Web site address for a Web site specially developed for the exhibition, whose link could be tapped to display the website on a viewer's device or downloaded for later use. It is important to note that, because this is a technology that works with one's smart, hand-held device, there are no special headsets needed to enter this world of AR. In this instance, the use of the app could bring additional traffic to the library and provide an even greater level of interest in the exhibition's subject. Armed with a strong level of curiosity, users might then choose to move on to other supplementary materials such as books, articles and videos on the topic, thus expanding the interest in a topic spurred on by the AR exhibit. In incorporating the vast array of materials available in multiple media can serve to support the obvious link to information literacy, where the user can build his or her skills and gain:

> […] the ability to know when there is a need for information, to be able to identify, locate, evaluate, and effectively use that information for the issue or problem at hand (NFIL, 2015).

Example 2 – EON reality

Another advantage of a VR or AR platform is that it provides the developer with a set of tools wherein original content can be created. No longer does the developer need to be satisfied with merely searching through and selecting content from a pre-determined menu of offerings. It is this shift to developing custom content that offers libraries the ability to build VR or AR experiences themselves. The developer need not be a professional. The ease of use of a number of these products places the power to conceive original content squarely in the hands of any librarian or educator, regardless of previous level of experience.

An example of an interactive VR solution through a three-dimensional experience may be found at EON Reality, where mobile VR (EON Mobile) can be selected or developed also for iOS or Android smart devices by libraries, educators and students. Currently in use at Carnegie Mellon University (USA), Imperial College (UK), Nanyang Technological University (Singapore) and many other educational institutions, where, in a blended learning environment, the combination of in-class and online learning

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provides fertile ground for an immersive VR experience that allows an educator or librarian to:

> […] combine 3D content with videos, sound effects, annotations, Wikipedia, PowerPoint, YouTube, and […] to upload their work to the EON Experience portal, an interactive online library that is home to thousands of 3D objects, avatars, scenes, and applications (EON Reality, 2015).

Example 3 – Layar

Based in Amsterdam, Layar is an augmented reality app also available for IOS, Android and Blackberry devices. Similar to the concept offered by Aurasma, the app allows creators to add multimedia of all types to bring static images to life and enhance the viewer's experience.

> With the Layar Creator, you can enhance flyers, postcards, packaging or any other item with interactive content, including video messages, Web and social links, photo slideshows, music clips […] (Layar, 2015).

VR/AR as a learning solution for "Digital natives"

> The mission of teachers in today's classroom is to create the proper conditions to determine the shift from the static transmission of knowledge to the student-centered learning. Students are now active participants in the learning process. In order to be successful and to ensure the engagement of students, teachers must use adequate methods common to the current generation of "digital natives" (Nanu *et al.*, 2013).

This concept, of course, is similar to the teaching and learning process in the library as it is for the pure academic process at a university or college. It speaks to student expectation and whether VR enhancements are developed for a course or a library experience, there can be a heightened level of knowledge and understanding of the virtual world and its electronic offerings brought to a generation, many of whom were raised on electronic devices and all of the enrichments that technology can provide. Because the basic understanding of these worlds is common to this audience, libraries are keenly aware of the importance of introducing all modes of material to support the work of the student. Librarians are always on the cutting edge of these technologies to engage their customers and using such tools to encourage higher levels of information literacy through all of the combined tools available to them is critical to student success.

Summary

VR and AR are not gimmicks, and must be seen as valid additions to the toolkit that may be used by libraries to engage its audience, not only with the latest technology but also with the goal in mind of ensuring a proper approach to teaching information literacy. Students will gain immeasurably from the enhanced delivery of information on a particular topic through VR or AR and the multiple means by which the student can become proficient in the basic information literacy skills culminating is successful search for information, using every tool at his or her disposal to complete their academic assignments. Using these tools and through their use, VR and AR technology enhancements offer librarians another opportunity to fit into the role of:

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[…] guides (where), visitors can time-travel to 1906 to witness the massive earthquake that tore apart San Francisco, or ride on the Pequod, chasing Moby Dick – gaining exposure to great books by entering and living in them (Sarno, 2014).

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